

2025 AMC 10A Problem 18 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 18. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 18

Problem:

The harmonic mean of a collection of numbers is the reciprocal of the arithmetic mean of the reciprocals of the numbers in the collection. For example, the harmonic mean of 4, 4, 5 is

$$\frac{1}{\frac{1}{3} \left(\frac{1}{4} + \frac{1}{4} + \frac{1}{5} \right)} = \frac{30}{7}.$$

What is the harmonic mean of all the real roots of the 4050th degree polynomial

$$\prod_{k=1}^{2025} (kx^2 - 4x - 3)?$$

Answer Choices:

A. $-\frac{5}{3}$

B. $-\frac{3}{2}$

C. $-\frac{6}{5}$

D. $-\frac{5}{6}$

E. $-\frac{2}{3}$

Solution:

Let

$$P(x) = \prod_{k=1}^{2025} (kx^2 - 4x - 3)$$

In order to compute the harmonic mean of all the roots of the polynomial $P(x)$, we need to find the sum of the reciprocals of all the roots of $P(x)$.

We observe that each quadratic factor in the polynomial $P(x)$ is $kx^2 - 4x - 3$. Let its roots be $r_{k,1}$ and $r_{k,2}$.

For any quadratic $ax^2 + bx + c = 0$, we know:

$$r_1 + r_2 = -\frac{b}{a} \quad \text{and} \quad r_1 r_2 = \frac{c}{a}.$$

Thus,

$$\frac{1}{r_1} + \frac{1}{r_2} = \frac{r_1 + r_2}{r_1 r_2} = \frac{-b/a}{c/a} = \frac{-b}{c}.$$

For the quadratic $kx^2 - 4x - 3$, here $a = k$, $b = -4$, $c = -3$, therefore

$$\frac{1}{r_{k,1}} + \frac{1}{r_{k,2}} = \frac{-(-4)}{-3} = -\frac{4}{3}.$$

Since there are 2025 such quadratics, the total number of roots is 4050, and the sum of the reciprocals of all roots is

$$\sum \frac{1}{r} = 2025 \left(-\frac{4}{3} \right) = -2700.$$

The harmonic mean H of all 4050 roots is given by

$$H = \frac{4050}{\sum \frac{1}{r}} = \frac{4050}{-2700} = \boxed{\text{(B)} -\frac{3}{2}}.$$

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